Reliability & Self-Healing Workflows

How Background Jobs Survive Server Reboots and Fix Their Own Mistakes

Resilience by Design: Surviving Crashes and API Failures

In production environments, network timeouts, API rate limits, and server restarts are inevitable. Fathom's durable job engine is designed to ensure that **no work is ever lost**.

1. DETACHED PROCESS EXECUTION (SETSID)

Background jobs run as independent operating system processes detached from the terminal session. Closing your laptop, terminating SSH, or exiting the CLI does not interrupt ongoing worker execution.

2. SELF-HEALING TASK AUGMENTATION

When a job fails (e.g. hitting an unexpected rate limit), the runner does not execute a blind restart. Attempt #2 automatically injects the previous error trace and partial files, prompting the agent to adapt its strategy and self-heal.

The Self-Correction Loop in Action

`Attempt 1` **Initial Execution Fails**
Worker attempts to scrape 50 pages using primary API; hits HTTP 429 Rate Limit after saving 20 records.

`Attempt 2` **Augmented Prompt & Recovery**
Worker inspects the 20 saved records in workspace, switches search backend to DuckDuckGo fallback, applies request throttling, and completes remaining 30 records.

STALE PROCESS DETECTION & RECOVERY

If a server experiences an abrupt power outage, Fathom's SQLite registry identifies dead PIDs upon reboot and marks interrupted jobs as `stale`, allowing 1-click seamless restarts without corrupted state.

Autonomous Multi-Hour Durability: Deploy complex, multi-stage research workflows with total confidence that temporary network glitches will be resolved automatically.